Minimum-CVaR Portfolios for a Rupee-Based Investor

Out-of-Sample, Scenario and Crisis Evidence

Arjun Prashanth*

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Abstract. I compare five portfolio construction rules, equal weighting, minimum variance, maximum Sharpe ratio, risk parity and minimum Conditional Value-at-Risk (CVaR), on twelve assets spanning Indian equity, global equity, gold, bonds and Bitcoin, with every price measured in rupees. Minimum CVaR is solved as a linear program. In a walk-forward backtest with 38 quarterly rebalances net of costs, equal weighting has the highest Sharpe ratio, 1.245, but none of its differences from the optimised rules is significant at the 5% level. The minimum-CVaR portfolio has the smallest maximum drawdown, 12.19%, and its weights barely change when it is re-estimated on 20,000 bootstrap and Student-t scenarios. In a replay of the 2008 crisis it gains +5.13% while equal weighting loses 18.45%, mainly because the rupee's depreciation raised the value of its dollar bonds. For a rupee investor, the currency in which returns are measured is part of the portfolio's diversification.

Keywords: conditional value-at-risk; expected shortfall; portfolio optimisation; walk-forward backtest; estimation error; currency risk; India
JEL classification: C61, D81, G11, G15

1. Introduction

Mean-variance optimisation (Markowitz, 1952) measures risk by the variance of portfolio returns. Variance treats gains and losses symmetrically, and it is a complete description of risk only when returns are normally distributed. Daily asset returns are not normal. In the sample used here every asset has positive excess kurtosis, 14.2 for an S&P 500 ETF and 11.1 for a Nifty 50 ETF, and the S&P 500 ETF's worst day, −12.59%, is about three times the worst day a normal distribution with the same mean and variance would produce in a sample of this length (−4.1%).

Conditional Value-at-Risk, the expected loss on the worst days, measures only the loss tail and responds to its shape. Rockafellar and Uryasev (2000) show that minimising CVaR over a sample of returns reduces to a linear program, which makes it a practical portfolio objective. Whether it is a good one out of sample is a separate question. Michaud (1989) argued that optimisers amplify estimation error, and DeMiguel et al. (2009) found that no sample-based optimised portfolio they examined consistently beat equal weighting out of sample across seven datasets.

This paper asks, for an investor who measures wealth in Indian rupees: does a portfolio that minimises historical CVaR deliver lower tail risk than the alternatives when it is estimated and held out of sample, and what does it cost in return? Two design decisions were made before any optimisation was run. Every rule would be evaluated in a walk-forward backtest that uses only past data, and an equally weighted portfolio would be included as a benchmark.

*Email: arjunprashanth6129@gmail.com. Code, data and the script that generates every number, table and figure in this paper are at https://github.com/arjunprashanth6129/cvar-portfolio-optimizer. An interactive summary of the results is at https://arjun-quant-projects.vercel.app/cvar.

The evidence supports the minimum-CVaR rule on its own objective and not on risk-adjusted return. Out of sample, equal weighting has the highest Sharpe ratio and minimum CVaR the fourth highest, though no difference between equal weighting and an optimised rule is statistically significant. Minimum CVaR has the smallest maximum drawdown, its Sharpe ratio improves the most between the in-sample and out-of-sample periods, and its solution barely changes across simulated scenario sets, including a stress set with far heavier tails than the data. Its strongest result, a gain through the 2008 crisis, comes mostly from the rupee: measured in dollars, the same portfolio would have experienced the rise in correlation that the 2008 literature documents.

The paper contributes evidence for a multi-asset portfolio measured in an emerging-market currency, where the exchange rate is a source of both risk and diversification. It combines three tests that are often reported separately, a walk-forward backtest with formal tests of Sharpe ratio differences, scenario re-estimation and a historical crisis replay, and it reports the conditions under which the main result would not hold. Section 2 defines the risk measures and the five rules, Section 3 the data and Section 4 the evaluation design. Section 5 presents the results and Section 6 the crisis replay. Sections 7 and 8 discuss the findings and their limitations, and Section 9 concludes.

2. Risk measures and portfolio rules

2.1 Value-at-Risk and Conditional Value-at-Risk

Let w be a vector of portfolio weights, r a vector of daily asset returns and $L = -w^\top r$ the one-day portfolio loss. At confidence level $\alpha = 0.95$, Value-at-Risk is the α-quantile of the loss distribution,

$$\mathrm{VaR}_\alpha(L) = \inf\{\ell : \Pr(L \le \ell) \ge \alpha\}, \tag{1}$$

the loss exceeded on a fraction $1 - \alpha$ of days. It says nothing about the size of losses beyond the threshold, and it is not subadditive, so the VaR of a combined portfolio can exceed the sum of the VaRs of its parts. Conditional Value-at-Risk is the expected loss beyond the threshold,

$$\mathrm{CVaR}_\alpha(L) = \mathbb{E}\left[L \mid L \ge \mathrm{VaR}_\alpha(L)\right], \tag{2}$$

also called expected shortfall. It is a coherent risk measure in the sense of Artzner et al. (1999), so it is subadditive and diversification never increases it (Acerbi and Tasche, 2002), and it is convex in the weights. The two measures can rank the same assets differently, as Section 3 shows for the assets used here.

2.2 Minimum CVaR as a linear program

The days that form the tail depend on the weights, so CVaR cannot be minimised by sorting the losses of a fixed portfolio. For N equally likely historical scenarios with losses $L_j(w) = -r_j^\top w$, Rockafellar and Uryasev (2000, 2002) show that

$$\mathrm{CVaR}_\alpha(w) = \min_\zeta \; F_\alpha(w, \zeta), \qquad F_\alpha(w, \zeta) = \zeta + \frac{1}{(1-\alpha)N} \sum_{j=1}^{N} \left(L_j(w) - \zeta\right)^+, \tag{3}$$

and that the minimising ζ is the VaR. Because each L_j is linear in w and the positive part of a linear function is convex, F_α is jointly convex in w and ζ. Replacing each positive part with an auxiliary variable gives the linear program

$$\min_{w,\,\zeta,\,u} \; \zeta + \frac{1}{(1-\alpha)N} \sum_{j=1}^{N} u_j \quad \text{s.t.} \quad u_j \ge -r_j^\top w - \zeta, \;\; u_j \ge 0, \;\; \mathbf{1}^\top w = 1, \;\; w \ge 0, \tag{4}$$

Table 1. The five portfolio construction rules

Rule	Objective	Inputs estimated
Equal weight	$w_i = 1/N$	none
Minimum variance	minimise $w^\top \Sigma w$	covariance matrix
Maximum Sharpe	maximise $(\mu^\top w - r_f)/\sqrt{w^\top \Sigma w}$	covariance matrix and expected returns
Risk parity	equalise $w_i(\Sigma w)_i$ across assets	covariance matrix
Minimum CVaR	minimise $\mathrm{CVaR}_{0.95}$ of daily loss	historical return sample

All portfolios are long-only and fully invested. μ and Σ are the sample mean and covariance of daily returns over the estimation window, and r_f is the risk-free rate. The rules are ordered by the inputs they estimate, from none to expected returns, which are estimated least precisely.

which is solved to its global optimum with CVXPY (Diamond and Boyd, 2016) and the Clarabel solver (Goulart and Chen, 2024). Adding the constraint $\mu^\top w \geq R$ for a grid of targets R traces the CVaR efficient frontier. Both results of Rockafellar and Uryasev are checked numerically on every run. For the full-sample minimum-CVaR portfolio, the optimal objective is 0.8361% against an empirical CVaR of 0.8356%, and 5.01% of days lie beyond the optimal ζ of 0.5575%, against a target of 5%.

2.3 The other four rules

Table 1 lists the five rules. All portfolios are long-only and fully invested. The maximum-Sharpe problem is not concave as stated. It is solved through the transformation of Charnes and Cooper (1962): minimise $y^\top \Sigma y$ subject to $(\mu - r_f)^\top y = 1$ and $y \geq 0$, then set $w = y/\mathbf{1}^\top y$. The risk-parity condition is a non-convex system when stated directly. It is solved in the form of Spinu (2013),

$$\min_{w>0} \ \tfrac{1}{2}\, w^\top \Sigma w - \tfrac{1}{N} \sum_{i=1}^{N} \log w_i, \tag{5}$$

whose normalised solution equalises the risk contributions $w_i(\Sigma w)_i$ (Maillard et al., 2010). The logarithmic barrier makes a zero weight infeasible, so risk parity holds every asset.

3. Data

The universe has twelve assets in five classes, chosen so that their large losses tend not to coincide (Table 2). Daily prices adjusted for splits and dividends were downloaded from Yahoo Finance for 5 January 2015 to 14 August 2026, giving 2,775 trading days with prices for every asset. The risk-free rate is 6% a year, appropriate for a rupee investor.

Figure 1 shows the two tail measures for each asset. CVaR ranges from 1.07% a day for US aggregate bonds to 9.84% for Bitcoin, and the ranking by CVaR does not follow the ranking by volatility. VaR and CVaR can also rank assets differently. The Indian gold ETF has a lower VaR than the Nifty 50 ETF, 1.244% against 1.410%, but the highest CVaR/VaR ratio of the twelve assets, 1.71: its losses beyond the threshold are large relative to the threshold itself. Every asset's realised CVaR exceeds that of a normal distribution with the same mean and variance, by a factor of 1.03 to 1.18 (Panel B), so a normal model fitted to the first two moments would understate the tail loss of every asset in the universe. US long-term Treasuries illustrate the difference between daily tail risk and drawdown: a moderate daily tail coexists with a large maximum drawdown, because drawdown depends on the sequence of returns and CVaR does not.

Table 2. Assets and their tail statistics, full sample

			Annualised (%)		Daily, 95% (%)					
Asset	Class	Quoted	Return	Vol.	VaR	CVaR	CVaR/VaR	vs normal	Ex. kurt.	Max DD (%)
US aggregate bond	Bonds	USD	5.7	8.1	0.74	1.07	1.45	1.04	7.7	11.4
Gold ETF (INR)	Gold	INR	16.0	15.8	1.24	2.12	1.71	1.07	16.3	24.4
US 20+ yr Treasury	Bonds	USD	2.7	16.3	1.59	2.19	1.38	1.04	4.7	43.9
Nifty 50 ETF	Indian equity	INR	11.4	14.9	1.41	2.22	1.58	1.18	11.1	36.3
Gold ETF (USD)	Gold	USD	16.4	17.6	1.61	2.49	1.54	1.12	5.3	23.8
World equity ETF	Global equity	USD	16.2	18.4	1.67	2.65	1.59	1.14	15.6	30.4
S&P 500 ETF	Global equity	USD	19.2	18.9	1.74	2.78	1.60	1.17	14.2	30.0
EM equity ETF	Global equity	USD	11.6	22.3	2.07	3.13	1.52	1.10	8.1	31.9
HDFC Bank	Indian equity	INR	11.6	22.7	1.99	3.20	1.61	1.10	9.4	41.1
Reliance	Indian equity	INR	19.2	27.3	2.32	3.58	1.54	1.03	8.2	45.1
Infosys	Indian equity	INR	11.1	27.2	2.43	3.84	1.58	1.10	7.9	48.2
Bitcoin	Crypto	USD	68.0	67.8	6.11	9.84	1.61	1.16	6.5	81.1

All statistics use daily returns in rupees from 5 January 2015 to 14 August 2026. VaR and CVaR are historical estimates at 95% for a single day, reported as positive losses. "vs normal" is the realised CVaR divided by the CVaR of a normal distribution with the same mean and variance. Assets are sorted by CVaR.

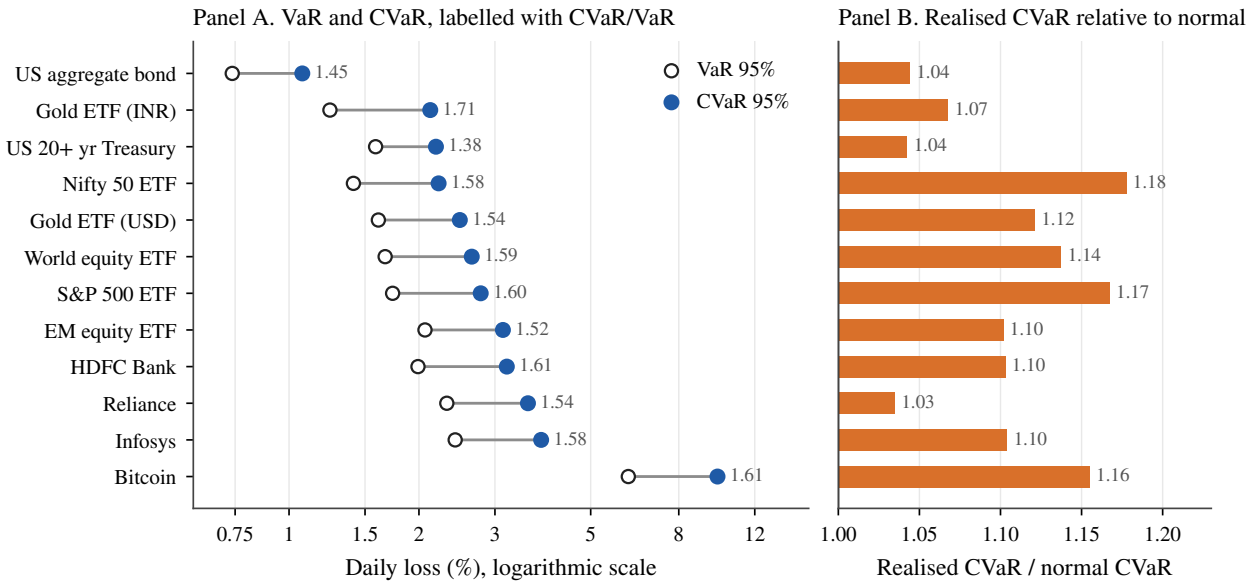


Figure 1. Tail risk by asset, from daily returns in rupees, 5 January 2015 to 14 August 2026. Panel A shows VaR at 95% (open circles) and CVaR at 95% (filled) on a logarithmic scale, so the length of each segment reflects the ratio CVaR/VaR, which is printed beside it. Panel B divides each asset's realised CVaR by the CVaR of a normal distribution with the same mean and variance. Assets are sorted by CVaR.

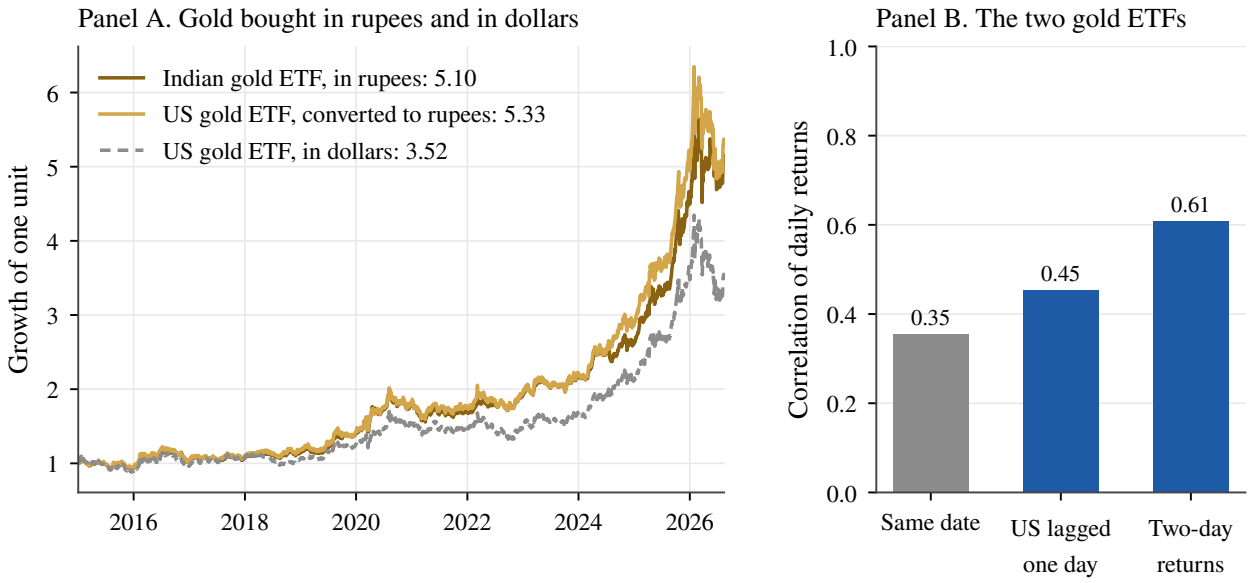


Figure 2. Checks on the currency conversion. Panel A shows the growth of one unit invested in the Indian gold ETF, priced in rupees, and in the US gold ETF, both converted to rupees at the daily exchange rate and in its own currency; the legend gives the final values. Panel B shows the correlation of the two ETFs' daily returns in rupees on the same date, with the US series lagged one day, and on overlapping two-day returns.

3.1 Currency conversion

Seven of the assets are quoted in US dollars. For a rupee investor, the return on a dollar asset combines its dollar return and the change in the exchange rate. Dollar prices are therefore multiplied by the daily USD/INR rate before returns are computed; converting prices rather than adding the two returns compounds them exactly. Over the sample the rupee moved from about 63 to about 95 per dollar. The exchange-rate series is committed with the code, so the restatement of the crisis windows in dollars in Section 6 is reproducible.

The conversion is checked by including gold twice: an Indian gold ETF priced in rupees and a US gold ETF priced in dollars. If the conversion is correct the two should end close together, and they do (Figure 2, Panel A). One rupee in the Indian ETF grows to 5.10 over the sample and one rupee in the converted US ETF to 5.33. Measured in dollars, the US ETF grows only to 3.52; the difference is the rupee's depreciation, which a rupee investor earns on every unhedged dollar asset. The remaining gap between the two rupee series has two identified sources, a difference in expense ratios and a fall in domestic gold prices when India cut its gold import duty in July 2024.

3.2 Asynchronous closing times

The NSE closes at 15:30 IST and the NYSE at 01:30 IST the following day, so Indian and US closes carrying the same date are about ten hours apart. Same-day correlations between the two markets are biased towards zero. For the two gold ETFs, which hold the same metal, the correlation of daily returns is 0.355 on the same date, 0.454 with the US series lagged one day and 0.608 on overlapping two-day returns (Figure 2, Panel B). All cross-market correlations in this paper are therefore understated, and the optimiser sees somewhat more diversification than exists. No correction is applied, because any correction would itself be a modelling assumption.

4. Evaluation design

4.1 In-sample and walk-forward evaluation

In-sample statistics are computed on the same data used to estimate the portfolios and are an upper bound on what an investor could have achieved. The out-of-sample test re-estimates every rule at each quarter-end using only the trailing 500 trading days, and holds the weights until the next rebalance (Figure 3). Between rebalances the weights drift with returns, and turnover at each rebalance is measured against the drifted weights. Costs of 20 basis points per unit of traded notional are deducted, so replacing the whole portfolio costs 40 basis points. This is intended to cover brokerage, securities transaction tax and stamp duty, the foreign-exchange spread and market impact for Indian large caps and US ETFs. There are 38 rebalances, and the out-of-sample period runs from 31 March 2017 to 14 August 2026, 2,240 trading days.

Annualised returns are compound returns. The Sharpe ratio is $(R_{\text{ann}} - r_f)/\sigma_{\text{ann}}$, with σ_{ann} the annualised standard deviation of daily returns. Differences in Sharpe ratios are tested with the test of Jobson and Korkie (1981) with the correction of Memmel (2003). For two return series with daily Sharpe ratios $\widehat{SR}_i$ and $\widehat{SR}_j$ estimated from T daily excess returns with correlation $\hat{\rho}$,

$$z = \frac{\widehat{SR}_i - \widehat{SR}_j}{\sqrt{\hat{\theta}}}, \qquad \hat{\theta} = \frac{1}{T}\left[2 - 2\hat{\rho} + \tfrac{1}{2}\left(\widehat{SR}_i^2 + \widehat{SR}_j^2 - 2\,\widehat{SR}_i\widehat{SR}_j\hat{\rho}^2\right)\right], \tag{6}$$

which is asymptotically standard normal under the null of equal Sharpe ratios.

4.2 Scenario sets

A CVaR estimated from history depends on the 139 days that form its 5% tail. To test how much the minimum-CVaR solution depends on those particular days, it is re-estimated on 20,000 scenarios from each of three generators.

Circular block bootstrap. Blocks of 10 consecutive days are drawn with replacement, wrapping at the end of the sample (Künsch, 1989; Politis and Romano, 1992). Resampling blocks preserves the clustering of volatility that resampling single days destroys: the first-order autocorrelation of squared returns is 0.291 in the data, 0.274 in the block bootstrap and −0.001 under single-day resampling.

Multivariate Student-t. A multivariate t distribution is fitted by the EM algorithm with a profile likelihood over the degrees of freedom. The estimate is $\nu = 5.90$, and the log-likelihood exceeds that of a multivariate

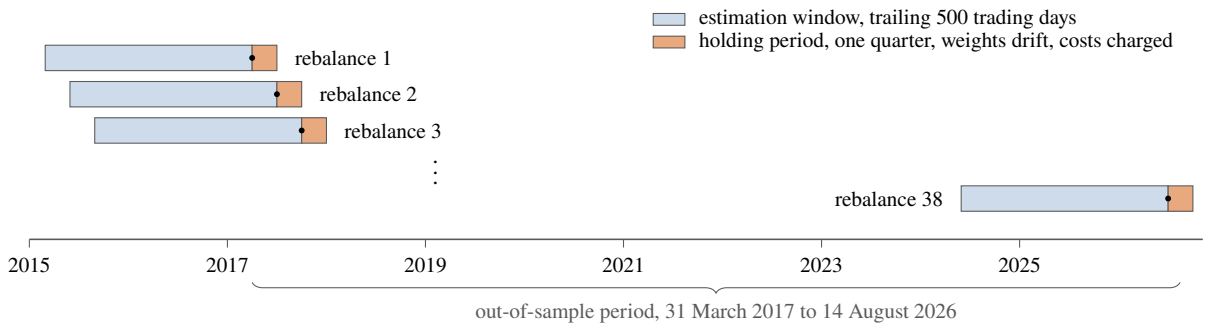


Figure 3. Walk-forward design. At each quarter-end (dots) every rule is re-estimated on the trailing 500 trading days (blue) and the resulting weights are held until the next quarter-end (orange), drifting with returns. Turnover is charged at 20 basis points per unit of traded notional. No estimation ever uses a day on which the portfolio is evaluated. The diagram is drawn to scale; only the first three and the last of the 38 rebalances are shown. Consecutive estimation windows overlap, and the holding periods tile the out-of-sample period without gaps.

normal by 2,831 over 2,775 observations. Sampling directly from the fitted Student-t model produced tails thinner than the data: maximum-likelihood estimation attributes extreme observations to the heavy tail rather than to the scale matrix, and the fitted scale implied only 95.6% of the sample volatility. The sampler is therefore rescaled to the sample covariance, so that only the tail shape differs from history. A further set with $\nu = 3$ is included as a stress case.

4.3 Crisis replay

The out-of-sample period contains the COVID-19 crash and a monetary tightening cycle but no credit crisis. The full-sample weights of each rule are therefore applied, unchanged, to the realised returns of earlier episodes. The main episode runs from 5 September 2008, shortly before the failure of Lehman Brothers, to the market low on 9 March 2009, 104 trading days. Three series did not exist for the whole window. The Nifty 50 ETF is replaced by the Nifty 50 index, which has a daily correlation of 0.971 with it where both exist, and the Indian gold ETF by the US gold ETF converted to rupees. Bitcoin did not exist, and no proxy is defensible, so two bounds are reported: Bitcoin removed with the remaining weights renormalised, and Bitcoin assumed to return three times the S&P 500.

Six further windows are replayed in the same way: the whole 2007-09 bear market (10 October 2007 to 9 March 2009), the worst month of 2008 (30 September to 27 October 2008), the 2011 euro-area crisis (25 July to 3 October 2011), the 2013 taper tantrum (23 May to 30 August 2013), the COVID-19 crash (20 February to 20 March 2020) and the 2022 rate rises (4 January to 14 October 2022). The last two lie inside the sample used to estimate the weights, so those replays are in sample; the other five precede it.

5. Results

5.1 In-sample and out-of-sample performance

Table 3 reports every rule in sample and out of sample. Comparing the two shows how much of each rule's in-sample performance came from estimating it on the evaluation data. Four of the five rules have a higher Sharpe ratio out of sample than in sample (Figure 4). The exception is maximum Sharpe, which falls from 1.388 to 0.832.

Maximum Sharpe is the only rule that uses estimated expected returns, and it is the rule most exposed to what Michaud (1989) called error maximisation. An optimiser assigns the largest weights to the assets whose expected returns are most overestimated in the sample, because it cannot distinguish overestimation from real attractiveness. Expected returns are estimated far less precisely than variances and covariances, so the effect is strongest for the rule that depends on them. Transaction costs do not explain the ranking. Maximum Sharpe turns over 25.4% of the portfolio per quarter, against 3.9% to 7.2% for the other rules, and loses 0.53 percentage points a year to costs, against at most 0.16 for the others, but the ordering by Sharpe ratio is the same before costs.

The never-rebalanced reference portfolio has the highest return but is not a neutral benchmark. Without rebalancing, an initially equal-weight portfolio drifts into the asset with the highest past return, and by the end of the period about 70% of its value is in Bitcoin, with volatility of 37.72% and a maximum drawdown of 57.61%.

5.2 Statistical significance

The out-of-sample period is 8.9 years long, which is short for comparing Sharpe ratios. Table 4 reports the test in equation (6). Equal weighting's advantage over the optimised rules has p-values between 0.086 and 0.272, and the difference between minimum CVaR and minimum variance has $p = 0.528$. None is significant

Table 3. In-sample and out-of-sample performance

	In sample, full period				Out of sample, walk-forward, net of costs					
Rule	Return	Vol.	CVaR	Sharpe	Return	Vol.	CVaR	Sharpe	Max DD	Turnover
Equal weight	19.03	11.68	1.625	1.115	21.21	12.21	1.710	**1.245**	21.72	4.7
Risk parity	14.26	8.70	1.189	0.949	15.68	9.01	1.243	1.075	17.28	4.0
Maximum Sharpe	24.83	13.56	1.895	1.388	19.33	16.04	2.389	0.832	34.36	25.4
Minimum CVaR	8.48	6.25	0.836	0.397	11.05	6.72	0.913	0.751	**12.19**	7.2
Minimum variance	8.73	6.22	0.842	0.439	10.66	6.55	0.882	0.712	12.72	3.9
Unoptimised references										
Nifty ETF, buy and hold					13.00	15.03	2.232	0.466	36.34	
Equal weight, never rebalanced					30.61	37.72	5.552	0.652	57.61	

Returns, volatility, maximum drawdown and turnover are in percent; CVaR is the daily historical CVaR at 95%, in percent. In-sample statistics use the full sample 5 January 2015 to 14 August 2026; returns there are annualised arithmetic means, the quantity the optimisers use. Out-of-sample statistics use the walk-forward backtest from 31 March 2017 to 14 August 2026, with compound annual returns, net of 20 basis points per unit of traded notional. Turnover is the average one-way turnover per rebalance. Rules are ordered by out-of-sample Sharpe ratio, and the best value in the two key columns is in bold. The Sharpe ratio uses $r_f = 6\%$.

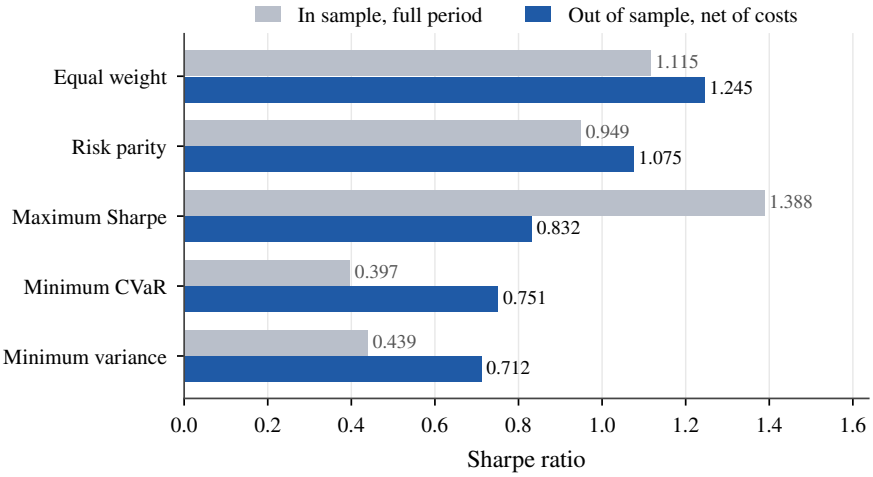


Figure 4. Sharpe ratio of each rule in sample, over the full period, and out of sample in the walk-forward backtest, net of costs. As in Table 3, in-sample ratios use arithmetic mean returns and out-of-sample ratios compound returns. Rules are ordered by the out-of-sample ratio.

at the 5% level. The data are consistent with equal weighting performing at least as well as the optimised rules, as DeMiguel et al. (2009) found more generally, but they do not establish that it performs better.

5.3 Drawdown and the accuracy of the tail-risk estimate

Figure 5 shows the cost of minimising tail risk and what it bought. One unit invested in the minimum-CVaR portfolio grows to 2.54 over the out-of-sample period, against 5.53 for equal weighting, because minimising tail risk concentrates the portfolio in low-volatility, low-return assets. In exchange, it has the smallest maximum drawdown of the five rules, 12.19%, against 21.72% for equal weighting and 34.36% for maximum Sharpe, which is close to the 36.34% drawdown of the Nifty ETF held alone.

Table 4. Tests of equal Sharpe ratios, out of sample

Rule i	Rule j	$\widehat{SR}_i$	$\widehat{SR}_j$	z	p-value
Equal weight	Risk parity	1.162	1.019	1.10	0.272
Equal weight	Maximum Sharpe	1.162	0.821	1.43	0.154
Equal weight	Minimum CVaR	1.162	0.730	1.55	0.120
Equal weight	Minimum variance	1.162	0.694	1.72	0.086
Minimum CVaR	Minimum variance	0.730	0.694	0.63	0.528

Jobson-Korkie test with Memmel's correction, equation (6), on daily excess returns net of costs over the walk-forward period, with the 6% annual risk-free rate converted to a daily rate by compounding. $\widehat{SR}$ is the annualised Sharpe ratio of daily arithmetic excess returns, which is lower than the ratio based on compound returns in Table 3. p-values are two-sided.

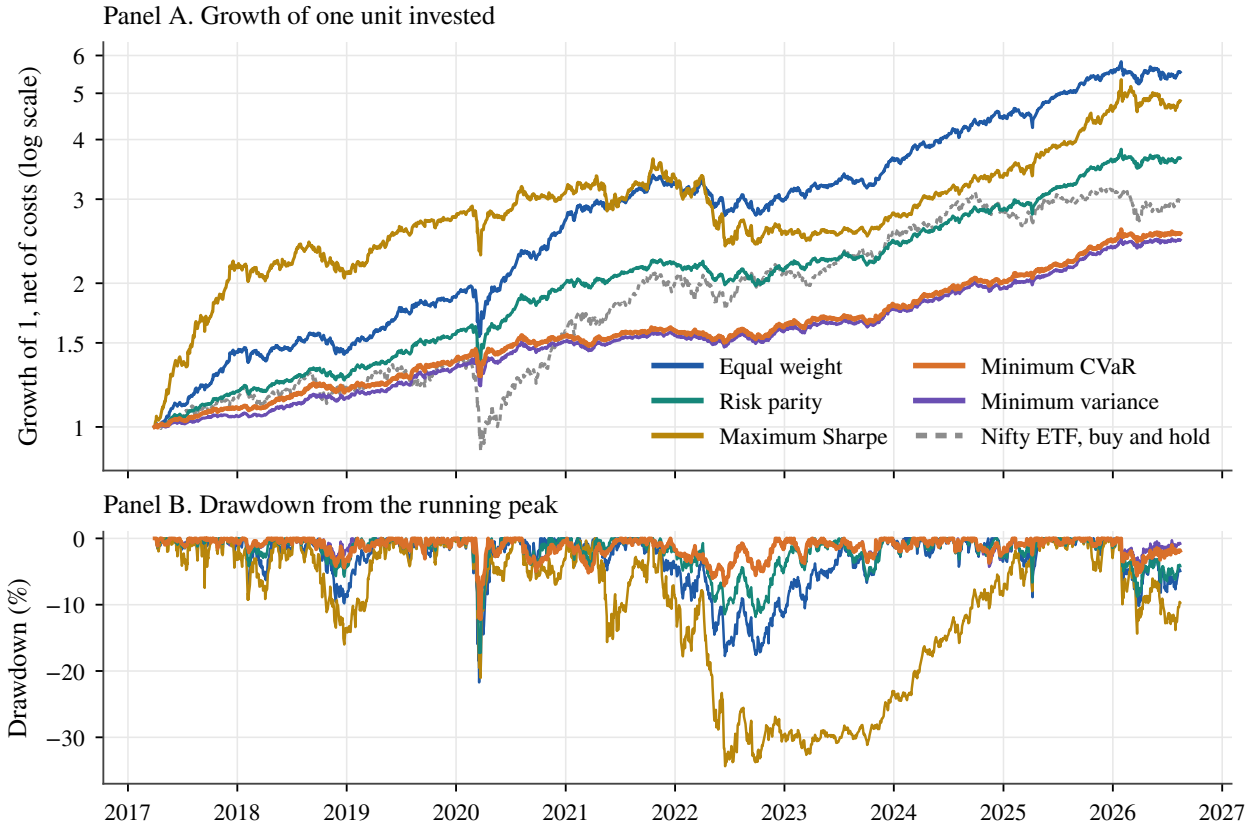


Figure 5. Out-of-sample performance, 31 March 2017 to 14 August 2026. Panel A shows the growth of one unit invested in each rule, net of costs, on a logarithmic scale, with the Nifty 50 ETF held throughout for reference. Panel B shows each rule's drawdown from its running peak.

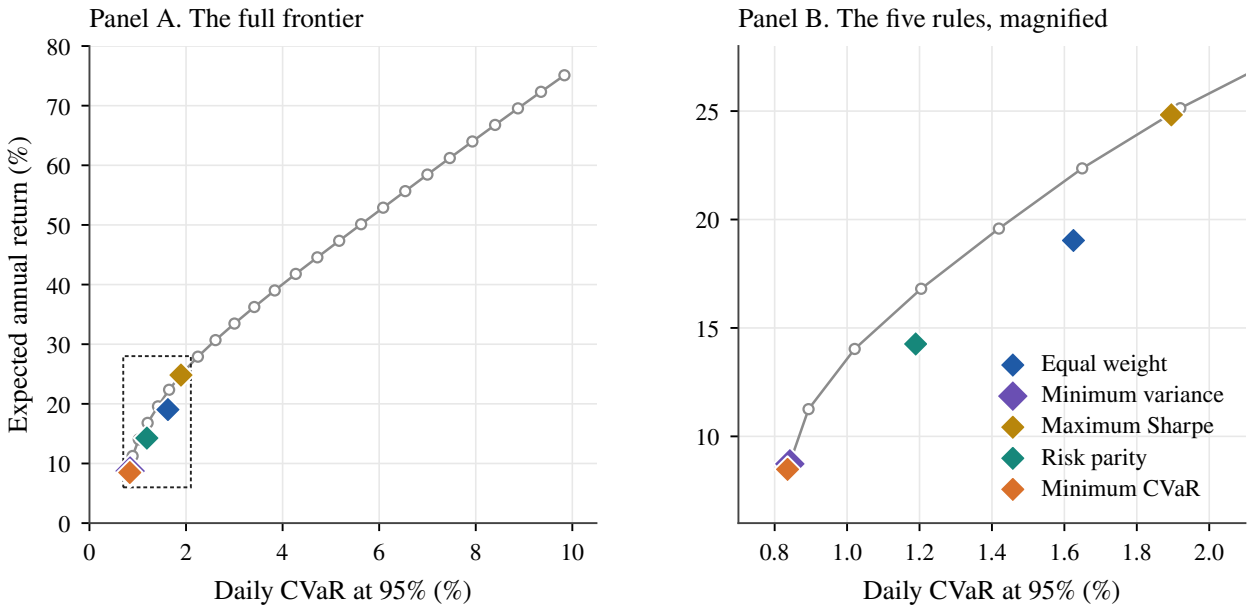


Figure 6. The CVaR efficient frontier, estimated on the full sample. Open circles are frontier portfolios and diamonds the five rules. Panel B magnifies the dashed region of Panel A, which contains all five rules. The minimum-variance marker is drawn larger because it lies almost on top of the minimum-CVaR marker.

The minimum-CVaR portfolio's realised CVaR out of sample is 0.913%, 9% above its in-sample estimate of 0.836%. That is a modest degradation, much smaller than the 26% of maximum Sharpe, but it is not the smallest of the five: risk parity and minimum variance degrade by 4% and 5%, and minimum variance has the lowest realised CVaR, 0.882%. The two minimum-risk rules are close on every tail measure, which is consistent with how similar their weights are (Section 5.5). Minimum CVaR's Sharpe ratio improves by +0.354 between the two periods, the largest change of the five.

5.4 The CVaR efficient frontier

Figure 6 shows the frontier traced by 25 solutions of the program in equation (4) with increasing return targets. No long-only portfolio of the twelve assets lies above or to the left of it; portfolios below or to the right of it, such as equal weighting and risk parity, carry more tail risk than their expected return requires. The marginal cost of return rises along the curve. Raising the expected return from 9% to 18% raises CVaR by 0.45 percentage points, or 5.0 basis points per point of return; between 35% and 50% the cost is 15.8 basis points per point. The highest-return point holds only Bitcoin and has a maximum drawdown of 81.06%.

Figure 7 shows how the portfolio changes along the frontier. At the low end, an expected return of 8.5%, it is 67% US aggregate bonds, with the Nifty ETF and Indian gold making up most of the rest. Bonds leave the portfolio by an expected return of about 22%. Gold, in its two forms, peaks at 54% of the weight near 25%. Above that, gold and the S&P 500 ETF give way to Reliance, which reaches at most 33%, and to Bitcoin, which reaches half the weight at about 50% and is the whole portfolio at the top. The steepening of the frontier follows from this composition: beyond the middle of the curve, each further point of expected return has to come from Bitcoin, whose CVaR is more than twice that of any other asset.

5.5 Composition and sensitivity to the scenario distribution

Minimum variance and minimum CVaR each hold four assets, with roughly two thirds of the weight in US aggregate bonds, the asset with the smallest CVaR (Figure 8, Panel A). Their CVaRs differ by only 0.006

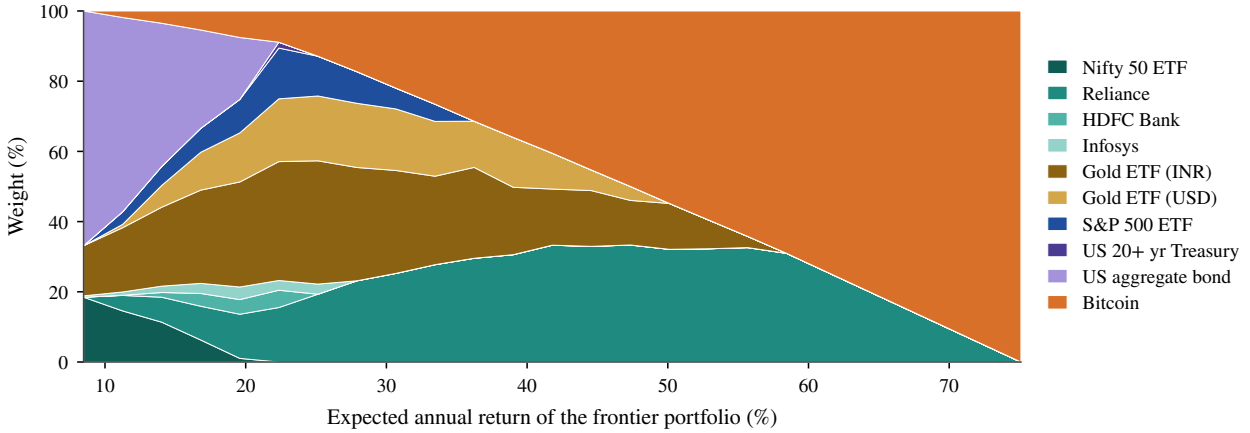


Figure 7. Composition of the CVaR efficient frontier. Each vertical slice is the minimum-CVaR portfolio for one expected-return target, from the lowest to the highest of the 25 points in Figure 6. Assets that never exceed 0.5% of a frontier portfolio are omitted.

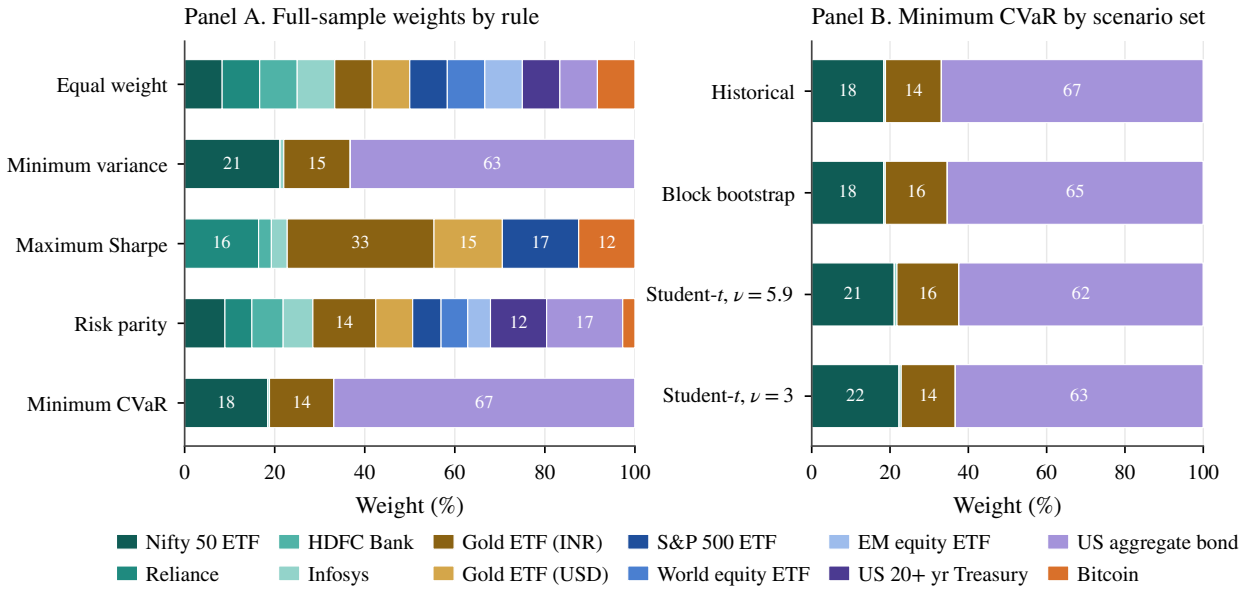


Figure 8. Portfolio weights. Panel A shows the full-sample weights of the five rules. Panel B shows the minimum-CVaR portfolio re-estimated on each scenario set; the Student-*t* sets are rescaled to the sample covariance. Labels give weights of 9% or more.

percentage points. Minimum CVaR moves about 3.7% of the weight from the Nifty ETF, gold and Infosys into bonds, and in sample this buys a smaller maximum drawdown, 11.00% against 11.32%, at slightly higher volatility, 6.25% against 6.22%. A variance objective cannot express that trade-off, because it does not distinguish the direction of deviations.

Maximum Sharpe is the only rule with a material Bitcoin allocation, 12.5%. Bitcoin's daily correlation with the four Indian equity assets is between 0.02 and 0.06, so a small allocation adds expected return with little effect on variance. Risk parity holds 2.7%: Bitcoin's volatility is about 3.6 times the average of the other assets, and equal risk contributions require a correspondingly smaller weight.

Table 5 and Figure 8, Panel B, show how the minimum-CVaR solution responds to the scenario sets. If its weights depended on the particular days in the historical tail, re-estimating them on different scenarios

Table 5. The minimum-CVaR solution across scenario sets

	Properties of the scenario set					Minimum-CVaR solution	
Scenario set	Vol. (%)	Ex. kurt.	Worst (%)	ACF r^2	Tail corr.	AGG (%)	Turnover (%)
Historical	23.1	9.6	−36.7	0.291	0.137	66.9	n/a
Circular block bootstrap	23.6	10.3	−36.7	0.274	0.166	65.3	1.60
Student-t, $\nu = 5.9$, scaled	23.3	6.9	−43.9	−0.002	0.022	62.4	4.46
Student-t, $\nu = 3$, scaled	23.3	71.4	−85.1	−0.002	0.148	63.3	4.02

Each set has 20,000 daily scenarios except the historical set, which has 2,775. Volatility, excess kurtosis and the worst single-asset daily return are averaged or taken across assets. ACF r^2 is the first-order autocorrelation of squared returns. Tail correlation is the average pairwise correlation of the twelve assets on the 5% of days with the largest equal-weight loss. AGG is the minimum-CVaR weight in US aggregate bonds, and turnover is measured against the historical solution.

would change them materially. They change little. Turnover relative to the historical solution is 1.60% for the block bootstrap, 4.46% for the fitted Student-t and 4.02% for the $\nu = 3$ stress set, whose mean excess kurtosis is 71.4 and whose worst single-asset day is −85.1%. The solution is determined by the structure of the assets, with US aggregate bonds having the smallest tail and the Nifty ETF and gold being the assets least correlated with them, more than by the specific days in the sample. Under the stress set the worst simulated day for minimum CVaR is −9.99%, against −28.02% for maximum Sharpe.

Both generators have known limits. The bootstrap cannot produce a loss larger than the largest historical loss; the worst single-asset day is −36.70% in both the historical and bootstrap sets. The Student-t understates the rise in correlation during crises. On the 5% of days with the largest equal-weight loss, the average pairwise correlation is 0.022 in the fitted Student-t scenarios against 0.137 in the data. An elliptical distribution produces extreme observations by scaling all assets by a common factor, which increases their size without aligning their signs, so diversification keeps working in its tails. Longin and Solnik (2001) find instead that correlations between equity markets rise in bear markets. Section 6 therefore uses historical crises directly.

6. Crisis replay and the currency effect

Table 6 reports the replay. In the 2008 window minimum CVaR gains +5.13% and minimum variance +2.96%, while equal weighting loses 18.45%. Neither minimum-risk rule holds Bitcoin, so the Bitcoin assumption does not affect them; it changes the maximum-Sharpe result by 12.2 percentage points. In the COVID-19 crash minimum CVaR loses 1.87% against 7.44% for equal weighting.

Figure 9 extends the replay to all 7 windows. Minimum CVaR has the highest return of the five rules in 6 of them; the exception is the 2022 rate rises, where minimum variance returns +3.88% against +3.64%. Equal weighting loses money in 6 of the 7 windows and minimum CVaR in 2. The 2013 taper tantrum is the only window in which every rule gains, from +12.0% for equal weighting to +13.7% for minimum CVaR, largely because the dollar rose 21.6% against the rupee and every rule holds dollar assets.

The 2008 result is not what the usual account of crises predicts. Average pairwise correlation across the eleven non-Bitcoin assets during the window was 0.159 in rupee terms, about 20% below its normal level of 0.199, measured the same way over the study period, rather than above it. The explanation is the exchange rate. Over the window the dollar rose 16.7% against the rupee, from 44.39 to 51.79 rupees, which raised the rupee value of every unhedged dollar asset while Indian equity prices were falling. The currency return offset part of the asset return and lowered measured correlation. The minimum-CVaR portfolio's gain comes mainly from its 67% allocation to dollar-denominated bonds during the depreciation.

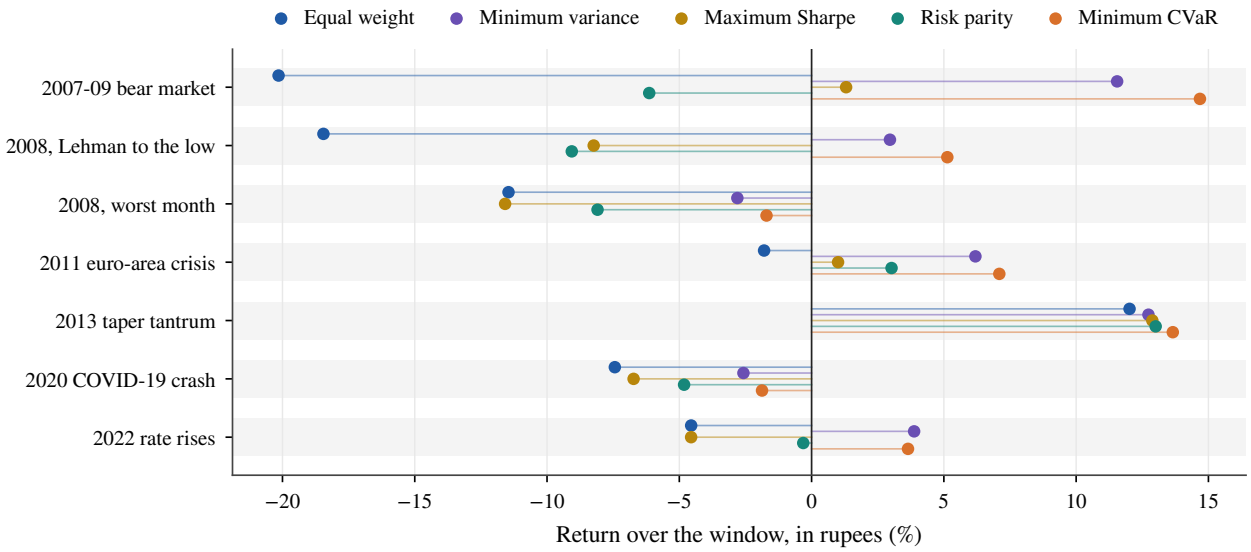


Figure 9. Return of each rule over the seven crisis windows, with full-sample weights applied unchanged to realised returns in rupees. Before Bitcoin's price history begins, it is removed and the other weights are renormalised. The 2020 and 2022 windows lie inside the estimation sample. Dates are given in Section 4.

Table 6. Historical crisis replay

	2008 return		2008	COVID-19	Bitcoin
Rule	Bitcoin dropped	Bitcoin = 3× S&P	max DD	return	weight
Equal weight	−18.45	−25.33	24.77	−7.44	8.3
Risk parity	−9.06	−11.73	16.47	−4.82	2.7
Maximum Sharpe	−8.24	−20.41	22.55	−6.73	12.5
Minimum variance	2.96	2.96	8.19	−2.58	0.0
Minimum CVaR	5.13	5.13	7.43	−1.87	0.0

Full-sample weights applied unchanged to realised returns in rupees. The 2008 window is 5 September 2008 to 9 March 2009. Bitcoin did not exist in 2008; the first column removes it and renormalises the other weights, and the second assumes it returned three times the S&P 500. Returns and maximum drawdown are in percent, and the Bitcoin weight is the rule's full-sample allocation.

Measuring the same window in dollars shows the effect directly. Average pairwise correlation is 0.159 in rupees and 0.254 in dollars, and the correlation between Indian equity and the US-listed assets, which determines whether the foreign holdings diversified the Indian ones, is 0.082 in rupees and 0.240 in dollars. The assets and days are identical; only the currency of measurement differs. In dollars, this portfolio experienced the rise in correlation documented for 2008. In rupees, the depreciation offset much of it.

Figure 10 repeats the comparison for every window. Average correlation is lower in rupees than in dollars in 5 of the 7. The gap is largest in the three windows from 2007-09: over the whole bear market, when the dollar rose 32.3% against the rupee, correlation was 0.142 in rupees against 0.213 in dollars. In 2011 and 2022 the two measures are almost equal. The two exceptions are the COVID-19 crash and the 2013 taper tantrum. In the COVID-19 crash the dollar rose only 5.0% against the rupee, too little to offset asset returns, and average correlation rose to 0.364 in rupees and 0.325 in dollars.

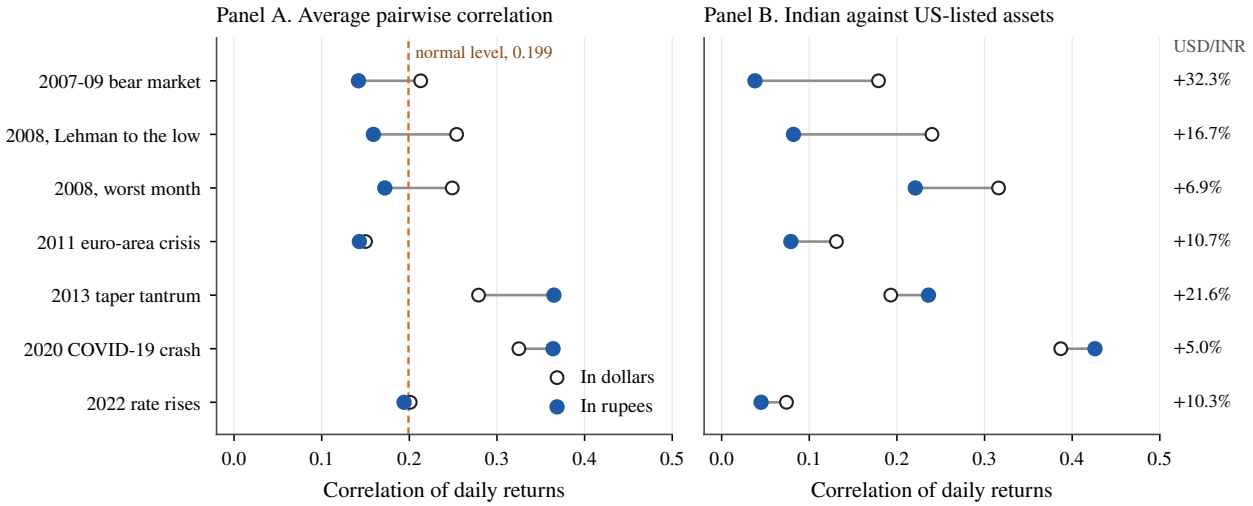


Figure 10. Correlation in each crisis window measured in rupees (filled) and in dollars (open). Panel A is the average pairwise correlation of daily returns across the assets with prices in the window. Panel B is the average correlation between the Indian assets and the US-listed assets. The column on the right gives the rise in the dollar against the rupee over each window. The dashed line is the normal level, the average pairwise correlation of the eleven assets other than Bitcoin, in rupees, over the study period; for the 2020 and 2022 windows the average includes Bitcoin, and the corresponding normal level is 0.186.

These episodes suggest a qualified rule for a rupee portfolio with unhedged foreign assets: correlations are dampened in global crises that come with a large rupee depreciation, and are not dampened otherwise. The 2013 taper episode does not fit it. The dollar rose 21.6% against the rupee, yet average correlation in rupees rose to 0.365, above both its normal level and the 0.279 measured in dollars. That episode was a funding shock to emerging markets rather than a global equity crash, and a large currency move adds a common component to every dollar asset measured in rupees, which raises their correlation with one another.

7. Discussion

The out-of-sample ranking follows the amount each rule estimates. Equal weighting estimates nothing and ranks first. Risk parity and the two minimum-risk rules estimate a covariance matrix or a return sample and improve out of sample. Maximum Sharpe, which also estimates expected returns, is the only rule whose Sharpe ratio falls. This is the pattern predicted by Michaud (1989) and documented by DeMiguel et al. (2009), although over 8.9 years the differences are not statistically significant.

Tail-risk minimisation and risk-adjusted return are different objectives, and the results reflect that. The minimum-CVaR portfolio met its objective out of sample: the smallest drawdown, a tail-risk estimate that degraded only modestly, a solution that barely changes across simulated scenarios, and the highest return of the five rules in 6 of the 7 crisis windows, including a gain in 2008. It also had the second-lowest return and a lower Sharpe ratio than equal weighting. Minimum variance came very close on every tail measure, which suggests that with this asset universe most of the protection comes from concentrating in low-risk assets, and the choice between the two objectives matters less than the choice to minimise risk at all.

For a rupee investor, diversification is a property of the assets and of the currency in which they are measured. The minimum-CVaR portfolio's best result depends on the rupee falling when global equities fell. That relationship held in 2008 and failed in 2013, so it should not be treated as a hedge that can be relied on.

8. Limitations

- *No credit crisis in the estimation window.* The out-of-sample period contains no event comparable to 2008. A longer sample adds tail observations, not a different kind of event.
- *Dependence on the exchange rate.* The 2008 result depends on the rupee depreciation, and no crisis without one has been tested.
- *Small tail sample.* CVaR at 95% averages about 139 observations, so it is much noisier than a volatility estimate.
- *Regime mixing.* A period of near-zero rates, a pandemic and a tightening cycle are combined in one covariance matrix and tail sample, and Bitcoin's return distribution changed substantially over the period.
- *Measurement.* Cross-market correlations are biased towards zero by asynchronous closes, all twelve assets were selected knowing they exist today, and the Bitcoin assumption for 2008 is arbitrary.
- *Parameters and costs.* The 500-day window and quarterly rebalancing are conventional choices made with knowledge of the data, and validating them would require a further holdout period. Costs exclude the market impact of large orders and capital gains tax.

9. Conclusion

A portfolio that minimises historical CVaR, estimated and held out of sample, delivered the protection it was designed for: the smallest drawdown of five rules, a modest gap between estimated and realised tail risk, weights that barely change across simulated scenarios, including a heavy-tailed stress set, and a gain through the 2008 crisis. It did not deliver the highest risk-adjusted return, which belonged to an equally weighted portfolio, although the difference is not statistically significant. Its crisis performance for a rupee investor rests largely on the exchange rate, which is a source of diversification that cannot be relied on in every crisis.

The most useful extensions follow from those limits: a currency-neutral crisis scenario, copula models that let correlations rise in the tails independently of the marginal distributions, regime-switching estimates of the covariance matrix and tail, and CVaR at several confidence levels.

Use of AI tools

The research question, the design of the study, the choice of portfolio rules and tests, and the interpretation of the results are my own. I used Claude Code, Anthropic's AI coding assistant, in two supporting roles. The first was as a tutor. When the work reached an idea I had not used before, such as how the Rockafellar-Uryasev reformulation turns the minimisation of CVaR into a linear program, I asked it to explain the idea and then checked the explanation against the data before relying on it. The second was implementation: at my direction it wrote parts of the code, including the plotting code and the script that generates this paper's numbers, tables and figures, and it helped me draft and edit the text. Working this way also taught me how to use an AI assistant well, by specifying a task precisely, asking for the reasoning behind a suggestion and not only the code, and verifying every output. I reviewed everything it produced and take full responsibility for the content.

Data and code availability

Prices, code and results are available in the project repository, https://github.com/arjunprashanth6129/cvar-portfolio-optimize
Running `python paper/make_assets.py` regenerates every number, table and figure in this paper from the committed output of the pipeline.

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